

University Fest Lab 7

PES2UG23CS285

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‘E’

Q1. List the names of participants who registered for events with a price above the average event price.

Command

```
SELECT DISTINCT p.name
FROM participant p
JOIN registration r ON p.SRN = r.SRN JOIN event e ON r.event_id = e.event_id
WHERE e.price > (SELECT AVG(price) FROM event);
```

Result Grid		Filter Rows:	Search	Export:
	name			
	Olivia Davis			
	Ethan Miller			
	Ava Wilson			
	William Taylor			
	Emma Anderson			
	Liam Martinez			
	Isabella Thompson			
	James Harris			
	Mia Davis			
	Benjamin Clark			
	Charlotte Baker			
	Daniel Lopez			
	Amelia Turner			
	Henry Hill			
	Victoria Young			
	John Smith			
	Sophia Brown			
	Jacob Jones			
	David Lee			
	Sofia Green			
	Christopher Moore			
	Scarlett Evans			
	Andrew Martinez			
	Emily Johnson			
	Michael Williams			

Q2. Show events that have more registrations than the average number of registrations per event.

Command

```
SELECT
    e.event_name,
    COUNT(r.SRN) AS number_of_registrations
FROM
    event e
JOIN
    registration r ON e.event_id = r.event_id
GROUP BY
    e.event_id, e.event_name
HAVING
    COUNT(r.SRN) > (
        SELECT AVG(registration_count)
        FROM (
            SELECT COUNT(SRN) AS registration_count
            FROM registration
            GROUP BY event_id
        ) AS event_counts
    );
```

Result Grid   Filter Rows: Search Export: 			
	event_name	number_of_registrati...	
	Music Concert	7	
	Art Workshop	10	
	Poetry Slam	10	
	Quiz Competition	11	
	Hackathon	8	
	Sound and Lighting	8	
	Event Coordination	10	
	Escape Room Challenge	11	
	Poetry Recitation	14	
	Code Jam	7	
	Tech Expo	8	
	Drama Play	11	

Task 2: Correlated Queries

Q3. Find participants who have registered for the highest-priced event(s)

Command

```
SELECT p.name
FROM participant p
WHERE p.SRN IN (
    SELECT r.SRN
    FROM registration r
    WHERE r.event_id IN (
        SELECT e.event_id
        FROM event e
        WHERE e.price = (SELECT MAX(price) FROM event)
    )
);
```

Result Grid   Filter		
	name	
	John Smith	
	Emily Johnson	
	Michael Williams	
	Sophia Brown	
	Jacob Jones	

Q4. Find stalls that offer items cheaper than the average price of that item across all stalls.

Command

```
SELECT
    s.name AS stall_name,
    si.item_name,
    si.price_per_unit AS offered_price,
    ROUND(avg_price_table.avg_price, 2) AS average_item_price
FROM
    stall_items si
JOIN
    stall s ON si.stall_id = s.stall_id
```

JOIN

```
(SELECT item_name, AVG(price_per_unit) AS avg_price  
FROM stall_items  
GROUP BY item_name) AS avg_price_table
```

ON

```
si.item_name = avg_price_table.item_name
```

WHERE

```
si.price_per_unit < avg_price_table.avg_price
```

ORDER BY

```
si.item_name, offered_price;
```

Result Grid   Filter Rows: Search Export: 					
	stall_name	item_name	offered_pri...	average_item_pri...	
	Culinary Haven	BBQ Chicken Sandwich	489.00	503.38	
	BiteStreet	BBQ Chicken Sandwich	490.00	503.38	
	Culinary Haven	Chicken Noodle Soup	519.75	546.38	
	BiteStreet	Chicken Noodle Soup	525.00	546.38	
	TasteBud Oasis	Chicken Noodle Soup	529.00	546.38	
	Savory Delights	Classic Caesar Salad	349.25	360.31	
	Flavor Fusion	Classic Caesar Salad	349.50	360.31	
	FlavorCraft	Classic Caesar Salad	360.00	360.31	
	TasteBud Oasis	Fish Tacos	449.00	462.93	
	YumBliss Café	Fish Tacos	450.00	462.93	
	BiteStreet	Fish Tacos	450.00	462.93	
	Culinary Haven	Grilled Steak	573.75	595.08	
	BiteStreet	Grilled Steak	589.00	595.08	
	FlavorCraft	Margherita Pizza	339.00	353.25	
	Savory Delights	Margherita Pizza	349.00	353.25	
	Flavor Fusion	Margherita Pizza	350.00	353.25	
	Munchie Magic	Mushroom Risotto	349.00	368.67	
	Flavor Fusion	Mushroom Risotto	359.00	368.67	
	Savory Delights	Mushroom Risotto	360.00	368.67	
	FlavorCraft	Mutton Stroganoff	559.00	576.75	
	SpiceHub	Shrimp Scampi	412.75	417.25	
	FlavorCraft	Spinach and Feta Om...	257.25	266.25	
	Flavor Fusion	Spinach and Feta Om...	259.00	266.25	
	Savory Delights	Vegetable Pad Thai	324.25	331.08	
	Flavor Fusion	Vegetable Pad Thai	329.00	331.08	
	Munchie Magic	Vegetable Stir-Fry	293.00	306.50	
	Flavor Fusion	Vegetable Stir-Fry	299.00	306.50	
	FlavorCraft	Vegetable Stir-Fry	300.00	306.50	
	Munchie Magic	Veggie Wrap	379.00	403.90	
	Flavor Fusion	Veggie Wrap	399.00	403.90	
	Savory Delights	Veggie Wrap	400.00	403.90	

Task 3: Window Functions

Q5. Show the SRN and name of participants and how many times they purchased something, along with their rank based on total quantity bought

Command

```
WITH ParticipantPurchaseSummary AS (  
  SELECT  
    p.SRN,  
    p.name,  
    COUNT(pu.timestamp) AS number_of_purchases,  
    SUM(pu.quantity) AS total_quantity_bought  
  FROM  
    participant p  
  JOIN  
    purchased pu ON p.SRN = pu.SRN  
  GROUP BY  
    p.SRN, p.name  
)  
SELECT  
  pps.SRN,  
  pps.name,  
  pps.number_of_purchases,  
  pps.total_quantity_bought,  
  DENSE_RANK() OVER (ORDER BY pps.total_quantity_bought DESC) AS purchase_rank  
FROM  
  ParticipantPurchaseSummary pps  
ORDER BY  
  purchase_rank, pps.name;
```

Result Grid  Filter Rows: Search Export: 						
	SRN	name	number_of_purchas...	total_quantity_bou...	purchase_rank	
	P1017	Daniel Lopez	7	22	1	
	P1005	Jacob Jones	9	21	2	
	P1002	Emily Johnson	4	12	3	
	P1003	Michael Williams	5	11	4	
	P1008	Ava Wilson	5	10	5	
	P1001	John Smith	4	10	5	
	P1010	Emma Anderson	5	8	6	
	P1024	Scarlett Evans	4	8	6	
	P1004	Sophia Brown	2	6	7	
	P1006	Olivia Davis	3	4	8	

Q6. Rank participants within their department based on the number of events they've registered for (should have 4 columns - department, name, total_number_of_events and rank).

Command

```
SELECT
  p.department,
  p.name,
  COUNT(r.event_id) AS total_number_of_events,
  DENSE_RANK() OVER (
    PARTITION BY p.department
    ORDER BY COUNT(r.event_id) DESC
  ) AS department_rank
FROM
  participant p
JOIN
  registration r ON p.SRN = r.SRN
GROUP BY
  p.SRN, p.name, p.department
ORDER BY
  p.department, department_rank;
```

departme...	name	total_number_of_eve...	department_ra...
Chemical	Andrew Martinez	13	1
Chemical	Jacob Jones	11	2
Chemical	Victoria Young	9	3
Chemical	Emma Anderson	7	4
Chemical	Benjamin Clark	5	5
Civil	Sophia Brown	11	1
Civil	Scarlett Evans	10	2
Civil	Henry Hill	7	3
Civil	William Taylor	7	3
Civil	Mia Davis	5	4
Computer...	John Smith	13	1
Computer...	David Lee	9	2
Computer...	Charlotte Baker	7	3
Computer...	Olivia Davis	7	3
Computer...	Liam Martinez	6	4
Electrical	Sofia Green	11	1
Electrical	Emily Johnson	10	2
Electrical	Daniel Lopez	8	3
Electrical	Ethan Miller	7	4
Electrical	Isabella Thomp...	6	5
Mechanical	Christopher Mo...	10	1
Mechanical	Michael Williams	9	2
Mechanical	Ava Wilson	7	3
Mechanical	Amelia Turner	7	3
Mechanical	James Harris	6	4

Task 4: Full Text Search

Q7. Ranked search: Order events by relevance to the word "Tournament"

Command

```
ALTER TABLE event ADD FULLTEXT(event_name);
SELECT
    event_name,
    MATCH(event_name) AGAINST('Tournament') AS relevance_score
FROM
    event
WHERE
    MATCH(event_name) AGAINST('Tournament')
ORDER BY
    relevance_score DESC;
```

Result Grid   Filter Rows: Search Export: 			
	event_name	relevance_score	
	Paintball Tournament	1.4499049186706543	
	Gaming Tournament	1.4499049186706543	

Q8. List stalls offering items whose names match "chicken" or "soup", and compute total stock per stall

Command

```
ALTER TABLE item ADD FULLTEXT(name);
SELECT
    s.name AS stall_name,
    SUM(si.total_quantity) AS total_stock
FROM
    stall s
JOIN
    stall_items si ON s.stall_id = si.stall_id
WHERE
    s.stall_id IN (
        SELECT DISTINCT stall_id
        FROM stall_items
        WHERE item_name IN (
            SELECT name
            FROM item
            WHERE MATCH(name) AGAINST('chicken soup');
```

```

        WHERE MATCH(name) AGAINST('chicken soup' IN NATURAL LANGUAGE MODE)
    )
)
GROUP BY
    s.stall_id, s.name
ORDER BY
    total_stock DESC;

```

Result Grid   Filter Rows: Search			
	stall_name	total_sto...	
	FlavorCraft	346	
	YumBliss Café	234	
	Munchie Magic	211	
	TasteBud Oasis	151	
	BiteStreet	125	
	Culinary Haven	120	
	SpiceHub	90	